

Creating a dynamic multi dimensional array?

```
TC
File Edit Run Compile Project Options Debug
Line 3 Col 38 Insert Indent Tab Fill Unindent * E
#include<alloc.h> /* <stdlib.h> */
#include<stdio.h> #include<conio.h>
void main()
{
int nr, nc, r, c, **p;
clrscr();
printf("Enter no of rows and columns ");
scanf("%d%d",&nr,&nc);
p = (int **)calloc(nr, sizeof(int));
for(r=0;r<nr;r++)p[r]=(int *)calloc(nc,sizeof(int));
printf("Enter %d integers\n",nr*nc);
for(r=0;r<nr;r++)for(c=0;c<nc;c++)scanf("%d",&p[r][c]);
puts("Elements are");
for(r=0;r<nr;r++)
{
for(c=0;c<nc;c++)
{printf("%3d",p[r][c]);}
printf("\n"); free(p[r]);p[r]=0;
}
free(p); p=NULL;
getch();
}
```

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F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run

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```
TC
Enter no of rows and columns 3
3
Enter 9 integers
1
2
3
4
5
6
7
8
9
Elements are
1 2 3
4 5 6
7 8 9
```

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```
TC
Enter no of rows and columns 3 2
Enter 6 integers
1 2 3 4 5 6
Elements are
1 2
3 4
5 6
```

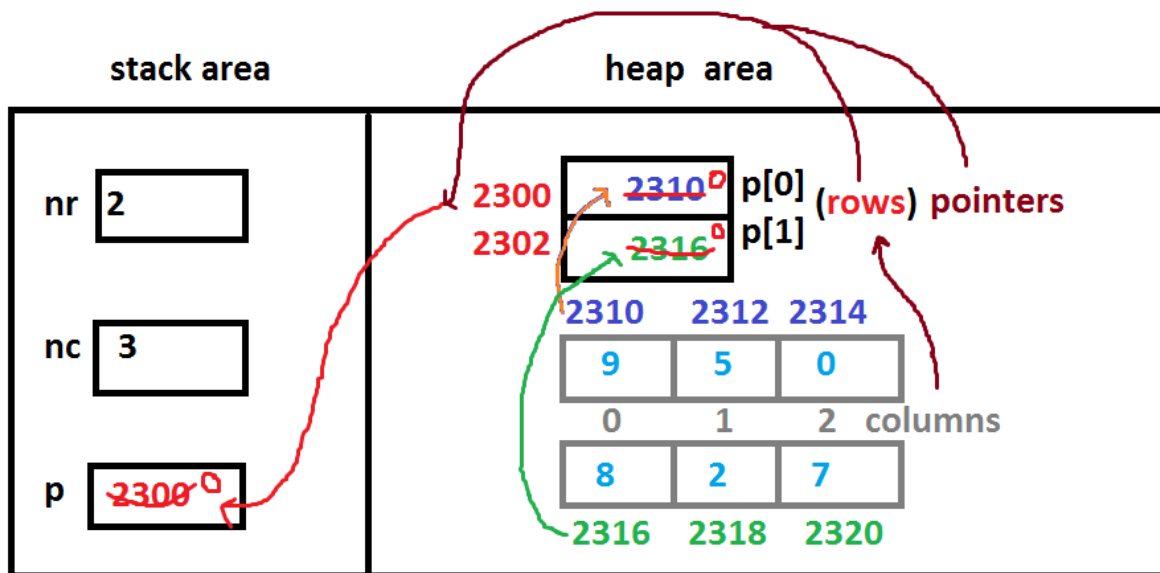
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Pointer style:

`scanf("%d", *(p+r)+c); /* &p[r][c] */`

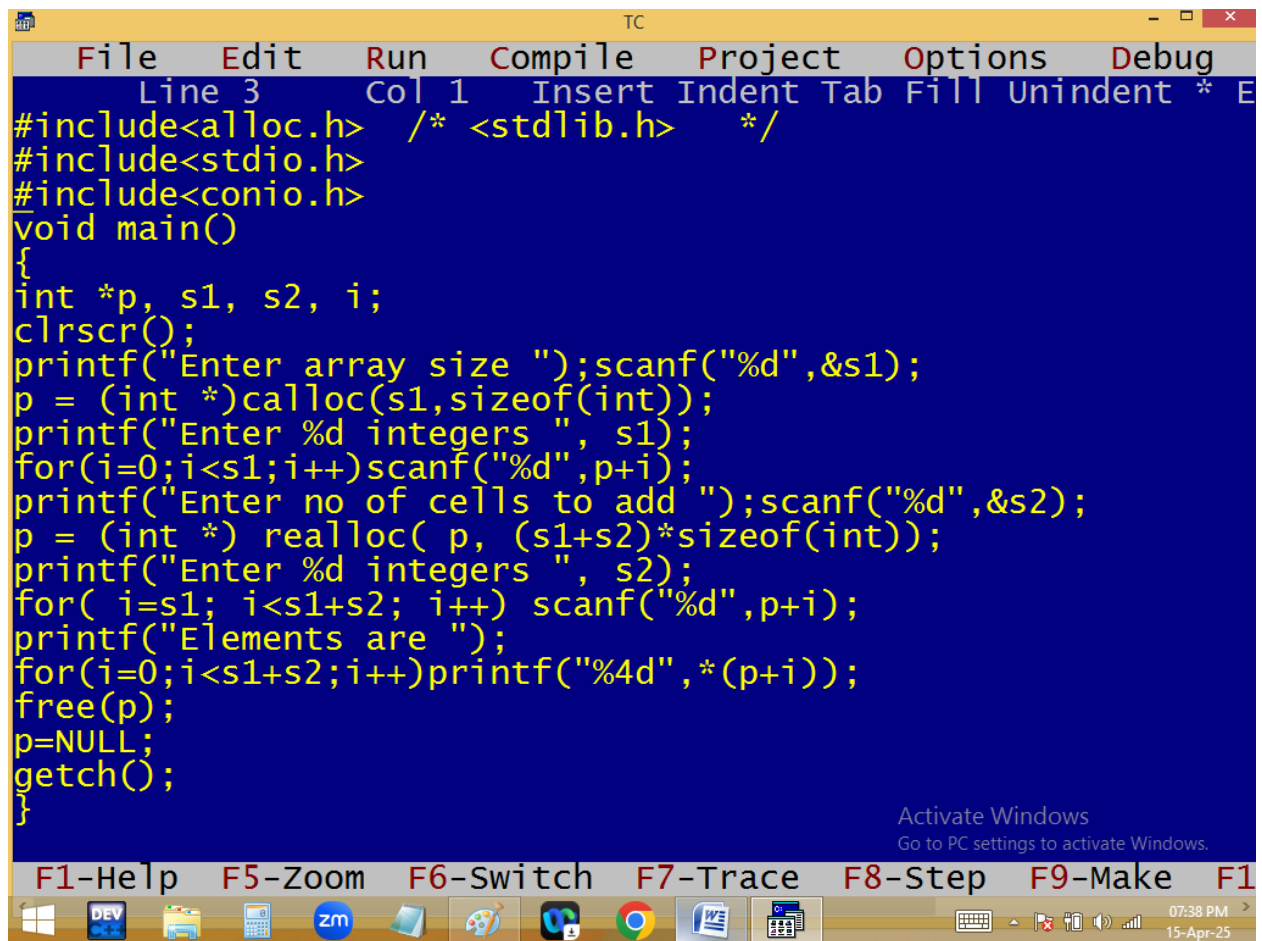
`Printf("%3d", *(*(p+r)+c) ); /* p[r][c] */`



realloc() example:

```
TC
File Edit Run Compile Project Options Debug
Line 3 Col 1 Insert Indent Tab Fill Unindent * E
#include<alloc.h> /* <stdlib.h> */
#include<stdio.h>
#include<conio.h>
void main()
{
int *p, s1, s2, i;
clrscr();
printf("Enter array size ");scanf("%d",&s1);
p = (int *)calloc(s1,sizeof(int));
printf("Enter %d integers ", s1);
for(i=0;i<s1;i++)scanf("%d",p+i);
printf("Enter no of cells to add ");scanf("%d",&s2);
p = (int *) realloc( p, (s1+s2)*sizeof(int));
printf("Enter %d integers ", s2);
for( i=s1; i<s1+s2; i++) scanf("%d",p+i);
printf("Elements are ");
for(i=0;i<s1+s2;i++)printf("%4d",*(p+i));
free(p);
p=NULL;
getch();
}

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```



```
TC
Enter array size 3
Enter 3 integers 2 7 4
Enter no of cells to add 2
Enter 2 integers 1 9
Elements are      2      7      4      1      9
```

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```
TC
Enter array size 2
Enter 2 integers 1 2
Enter no of cells to add 7
Enter 7 integers 3 4 5 6 7 8 9
Elements are 1 2 3 4 5 6 7 8 9
```

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stack area heap area

